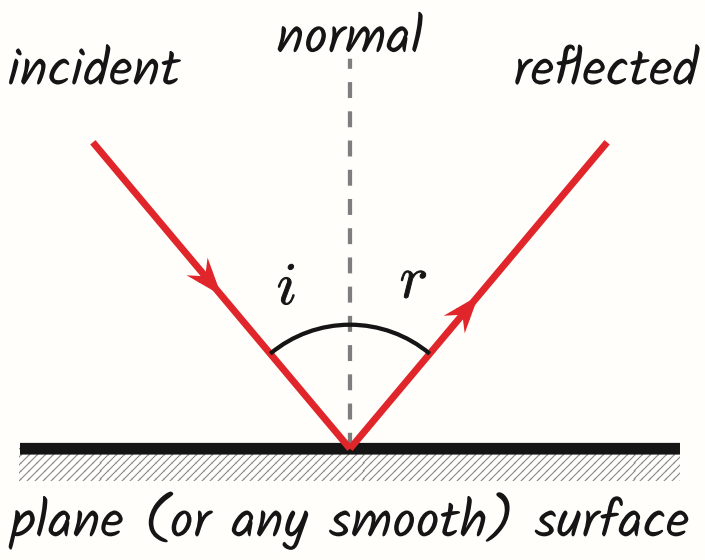
Reflection of Light

Class 12 · Cartesian sign convention, light from the LEFT · paraxial rays · thin lenses in air unless stated · examples: $n_{glass} = 1.5$, $n_{water} = 4/3$, $D = 25\,\text{cm}$

1 Laws of reflection



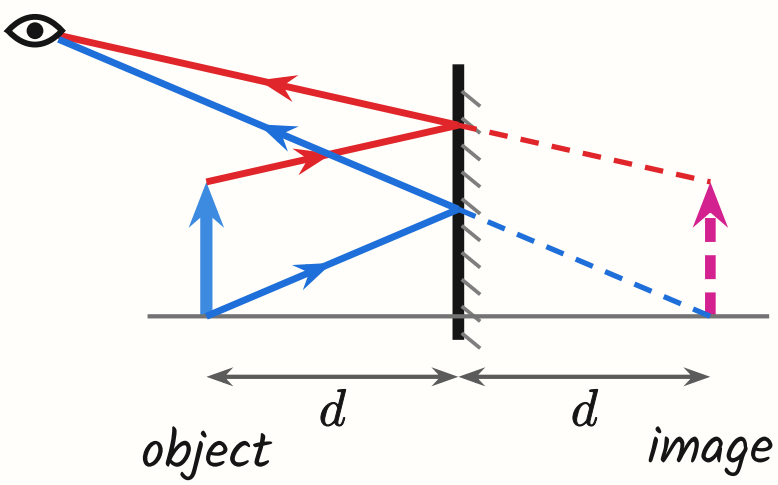
$$\angle i = \angle r$$

angles from the NORMAL · incident ray, normal, reflected ray in ONE plane · plane AND curved surfaces

$$\hat{e} = \hat{i} - 2(\hat{i} \cdot \hat{n})\,\hat{n}$$

$\hat{i}$, $\hat{e}$: along incident, reflected ray · $\hat{n}$: unit normal

2 Image in a plane mirror



- VIRTUAL, erect, same size, as far BEHIND as the object is in front ($m = +1$).
- Laterally inverted.
- PERFECT image with ALL rays (no paraxial condition).

3 Rectilinear propagation

- HOMOGENEOUS medium: light goes in straight lines; ray optics holds when apertures and obstacles $\gg \lambda$.

Syllabus Checklist

official JEE Main + JEE Advanced syllabus, NCERT, standard results

1 JEE Main + JEE Advanced syllabus

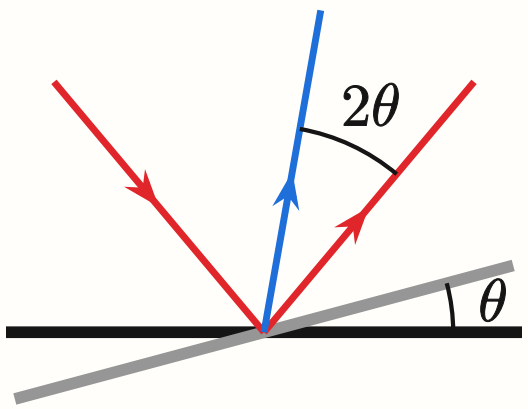
- ✓ Rectilinear propagation; reflection, plane mirrors — p1, p3
- ✓ Spherical mirrors: sign convention, mirror formula, m — p4–p7
- ✓ Refraction, Snell's law, refractive index — p8
- ✓ Apparent depth, normal shift, lateral shift — p9, p10
- ✓ Total internal reflection, critical angle, optical fibre — p11, p12
- ✓ Refraction at a spherical surface — p13
- ✓ Thin lens: lensmaker's formula, lens formula, m — p14–p16
- ✓ Power of a lens; lenses in contact and separated — p17
- ✓ Combinations of mirrors and thin lenses — p18
- ✓ Prism: deviation, minimum deviation, thin prism — p20, p21
- ✓ Dispersion, dispersive power — p22, p23
- ✓ Simple and compound microscope — p25, p26
- ✓ Astronomical (refracting) and reflecting telescopes — p27, p28

2 Also covered: NCERT / NEET + standard results

- ✓ Human eye and its defects (NCERT, NEET) — p24
- ✓ Inclined mirrors, rotating mirror, velocity of image — p3, p7, p16
- ✓ Newton's formula $x_1x_2 = f^2$, u–v graph — p7
- ✓ Lens in a liquid; different media on two sides — p14
- ✓ Silvered lens; displacement method; $4f$ minimum — p16, p18, p19
- ✓ Grazing incidence, no-emergence ($A > 2C$) — p21
- ✓ Achromatic, direct vision; scattering (NCERT, NEET) — p22, p23
- ✓ Terrestrial, Galilean, Newtonian and Cassegrain telescopes — p27, p28
- ✓ Traps, results table, solved questions — p29–p35

Plane Mirror Results

1 Rotating the mirror



reflected ray turns by 2θ

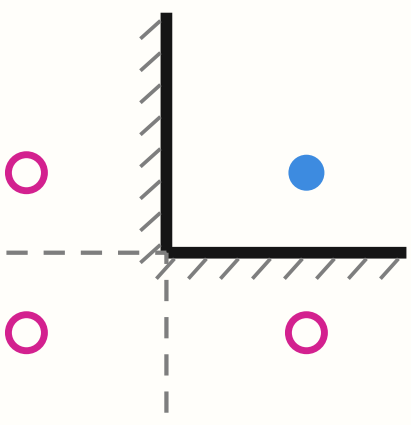
incident ray FIXED, mirror (grey) turned by θ

$$\delta = 180^\circ - 2i$$

deviation by ONE plane mirror

2 Two inclined mirrors

$\theta = 90^\circ$: *3 images*



$$n = \frac{360^\circ}{\theta} - 1$$

$360^\circ/\theta$ an EVEN integer · ODD: $n = \frac{360^\circ}{\theta}$ (object OFF the bisector) or $\frac{360^\circ}{\theta} - 1$ (ON it)

Example

$\theta = 60^\circ$: $n = 6 - 1 = 5$. $\theta = 72^\circ$: 5 (off the bisector) or 4 (on it). Parallel: infinitely many.

3 Velocity of the image

$$v_{I\parallel} = v_{O\parallel}, \qquad v_{I\perp} = 2v_{M\perp} - v_{O\perp}$$

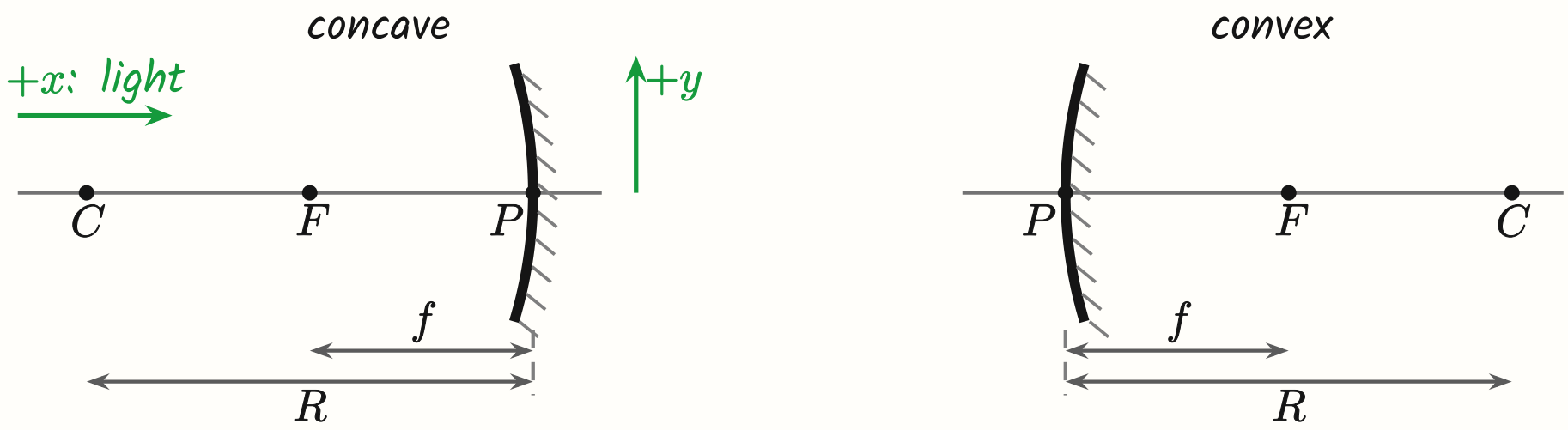
plane mirror, GROUND frame · $\parallel$ along it, $\perp$ along its normal · object at rest: image at $2v_M$

Golden points

★ *Full-length image (height H): mirror of $H/2$, any distance.*

Spherical Mirrors

1 Terms and Cartesian sign convention



$$f = \frac{R}{2}$$

PARAXIAL rays · distances from P, + along the incident light, heights + up · concave $f, R < 0$; convex $f, R > 0$ · MIRROR: f independent of the medium

2 Why only paraxial rays

$$CF' = \frac{R}{2\cos\theta} \rightarrow \frac{R}{2} \text{ only for small } \theta$$

parallel ray at height h, $\sin\theta = h/R$ · F': where it crosses the axis

Example

$h = R/2$: $\theta = 30°$, $CF' = 0.577R$, $PF' = 0.423R$ (not $0.5R$): wide rays focus closer (spherical aberration); a PARABOLIC mirror does not.

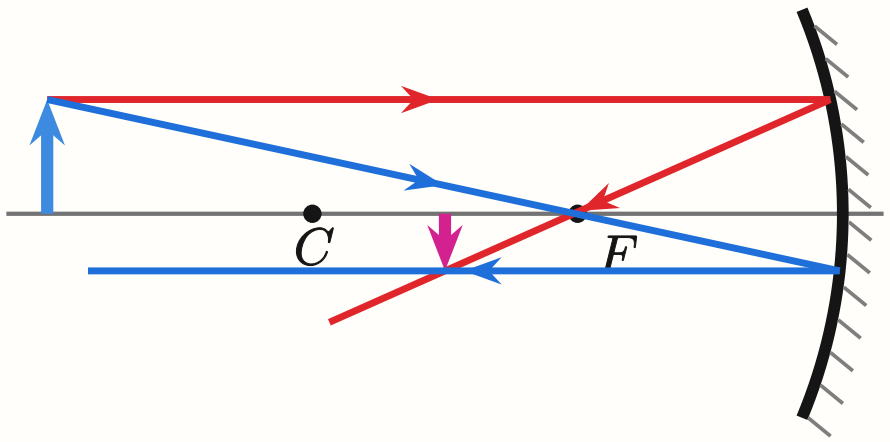
Golden points

★ Parallel $\rightarrow$ through F (or as if from F); towards $F \rightarrow$ parallel; towards $C \rightarrow$ back on itself; at $P \rightarrow$ symmetric about the axis (paraxial).

Mirror Ray Diagrams

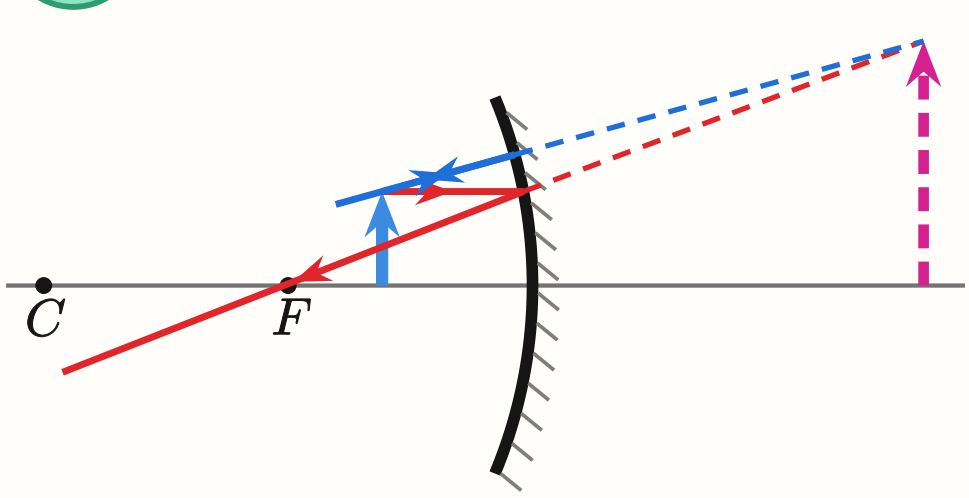
paraxial rays; heights exaggerated for clarity

1 Concave mirror, object beyond C



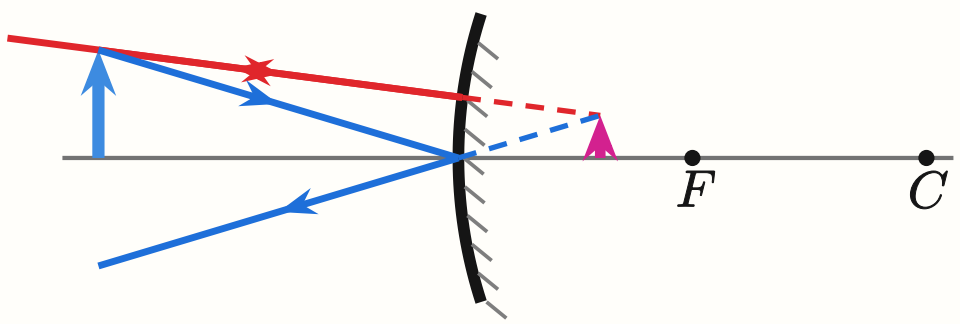
- Image between F and C: REAL, inverted, diminished.
- Red: parallel $\rightarrow$ through F. Blue: through $F \rightarrow$ parallel.

2 Concave mirror, object between F and P



- Image BEHIND the mirror: VIRTUAL, erect, magnified (shaving / make-up mirror).
- Blue: along $C \rightarrow$ object, returns on itself.

3 Convex mirror, any real object



- Image between P and F behind: VIRTUAL, erect, DIMINISHED (rear-view mirror: wide field).
- Red: towards C, back on itself. Blue: at P, symmetric.

Mirror Formula

1 Mirror formula and magnification

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}=\frac{2}{R}, \qquad m=\frac{h'}{h}=-\frac{v}{u}$$

PARAXIAL rays · Cartesian: from the pole, + along the incident light, heights + up · real object $u<0$ · concave and convex

2 Where is the image? (concave mirror)

Object	Image	Nature
at ∞	at F	real, inverted, point size
beyond C	between F and C	real, inverted, diminished
at C	at C	real, inverted, same size
C to F	beyond C	real, inverted, magnified
at F	at ∞	real, inverted, very large
F to P	behind the mirror	VIRTUAL, erect, magnified
convex: any	P to F, behind	VIRTUAL, erect, diminished

- Single mirror, real object: $m<0$ real, inverted, in FRONT ($v<0$); $m>0$ virtual, erect, BEHIND ($v>0$).

Example

Concave, $|f|=10\,\text{cm}$, object at $30\,\text{cm}$: $\frac{1}{v}=-\frac{1}{10}+\frac{1}{30}=-\frac{1}{15}$, $v=-15\,\text{cm}$ (real); $m=-\frac{-15}{-30}=-0.5$.

Image Velocity and Newton's Formula

1 Velocity of the image (spherical mirror)

$$v_I = -m^2\, v_O \quad \text{(along the axis)}$$

object moving ALONG the axis, mirror at rest, paraxial · velocities along $+x$ · image moves OPPOSITE to the object

$$v_{I\perp} = m\, v_{O\perp}$$

object moving PERPENDICULAR to the axis at fixed u

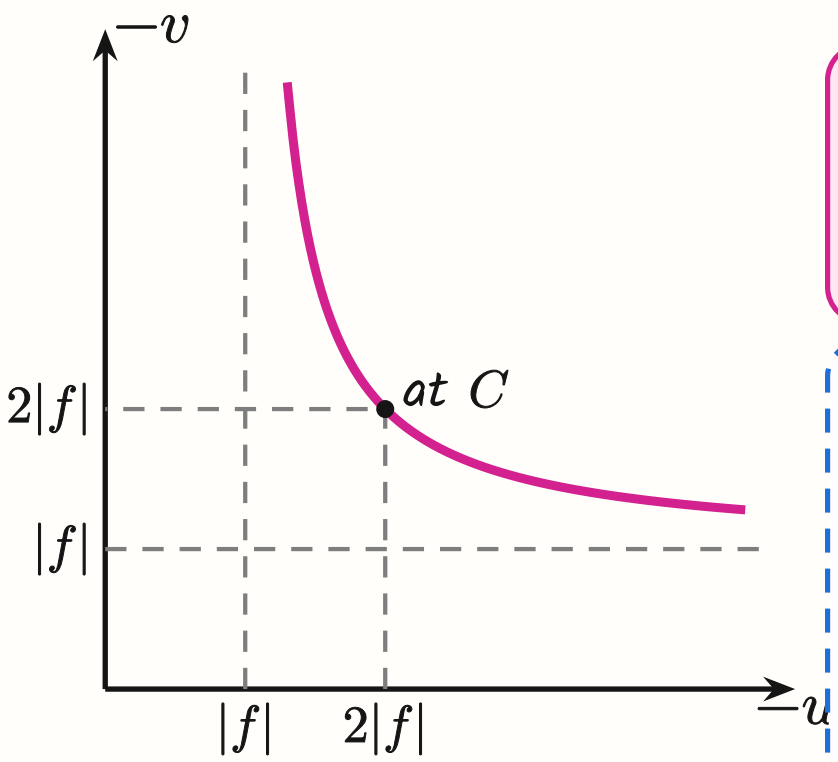
Example

Concave, $|f| = 20\,\text{cm}$, object at $60\,\text{cm}$ approaching at $3\,\text{cm}\,\text{s}^{-1}$: $v = -30\,\text{cm}$, $m = -0.5$, $v_I = -0.75\,\text{cm}\,\text{s}^{-1}$ *(moves AWAY).*

Golden points

★ *Plane mirror ($m = +1$, at rest): $v_I = -v_O$ along the normal. Object near F ($|m|$ large): the image moves very fast.*

2 Newton's formula and the u–v graph



concave mirror, real images

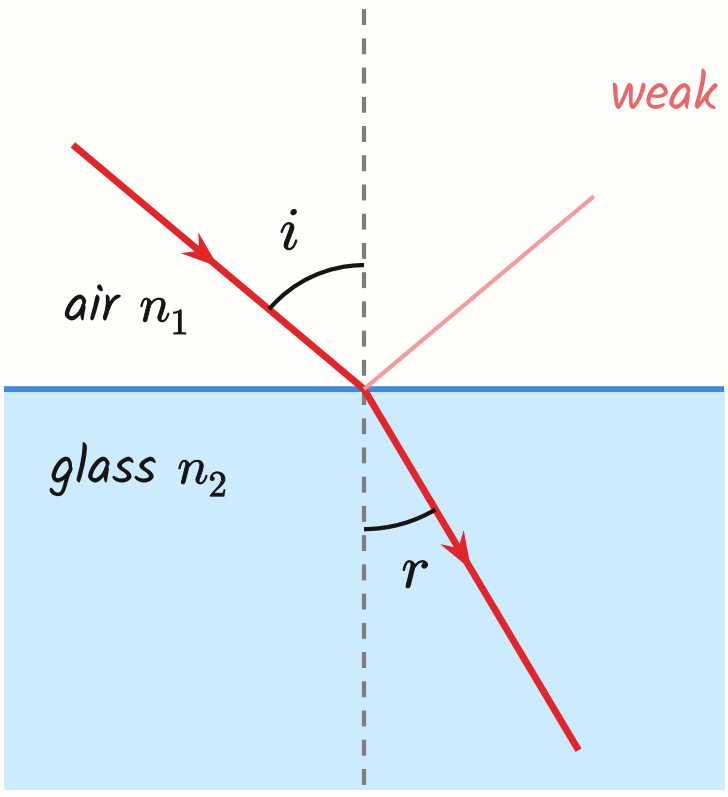
$$x_1 x_2 = f^2$$

x_1, x_2: distances of object and image from the FOCUS · paraxial, same for a thin lens

- *Graph: rectangular hyperbola, asymptotes at $|f|$, meets $v = u$ at $2|f|$.*
- $u = -30$, $f = -10$, $v = -15$: $x_1 = 20$, $x_2 = 5$, $x_1 x_2 = 100 = f^2$.

Refraction and Snell's Law

1 Snell's law



$$n_1 \sin i = n_2 \sin r$$

i, r from the NORMAL · incident ray, normal and refracted ray in one plane · $n = c/v$: absolute refractive index

Example

Air to glass ($n = 1.5$), $i = 50°$: $\sin r = 0.766/1.5$, $r = 30.7°$ (as drawn).

2 What changes, what does not

- FREQUENCY (colour) unchanged; $v = c/n$, $\lambda_m = \lambda_0/n$.
- Into a denser medium ($n_2 > n_1$): bends TOWARDS the normal; $i = 0$: no bending.
- Relative index $n_{21} = n_2/n_1 = v_1/v_2$; reversibility: $n_{12} = 1/n_{21}$, and $n_{12}\, n_{23}\, n_{31} = 1$.

Example

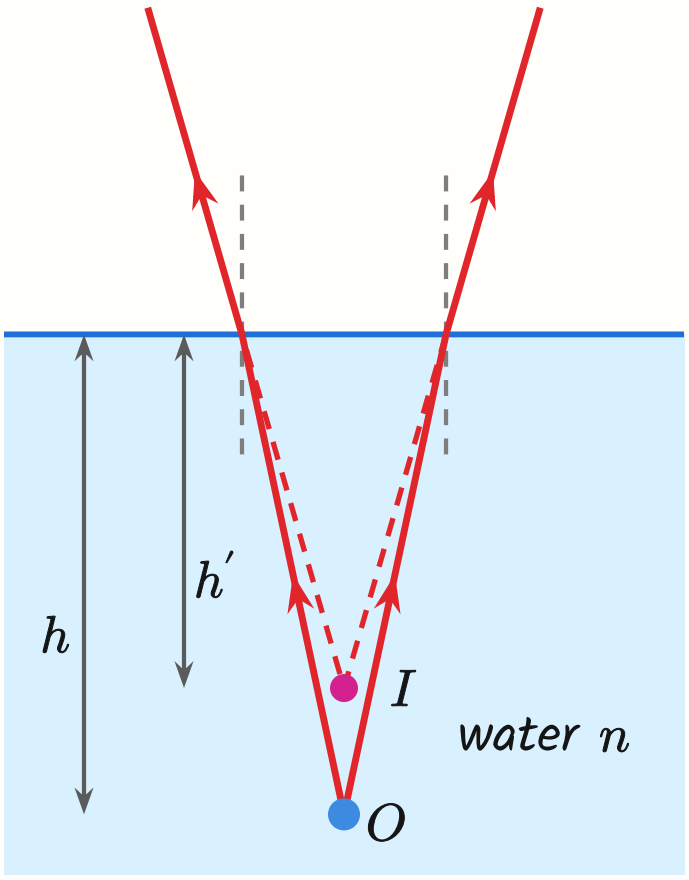
Glass, $n = 1.5$: $v = \frac{3\times10^8}{1.5} = 2 \times 10^8\,\mathrm{m\,s^{-1}}$; $600\,\mathrm{nm}$ (air) becomes $400\,\mathrm{nm}$ inside, frequency stays $5 \times 10^{14}\,\mathrm{Hz}$.

Golden points

★ Optical path $= nd$; time to cross thickness d along the normal: $t = nd/c$.

Apparent Depth and Normal Shift

1 Object under water, seen from above



$$h' = h\frac{n_{obs}}{n_{obj}} \Rightarrow h' = \frac{h}{n}$$

NEAR-NORMAL viewing · second form: object in medium n, observer in air

$$\Delta = h\left(1 - \frac{1}{n}\right)$$

rise of the image (near-normal viewing)

drawn exactly: $12°$ rays meet at $0.74h$; near-normal rays give $h/n = 0.75h$

2 Layers, slabs and the view from inside

$$h' = \frac{t_1}{n_1} + \frac{t_2}{n_2} + \cdots, \qquad \text{slab shift} = t\left(1 - \frac{1}{n}\right)$$

near-normal viewing from air · slab (t) between object and eye: image moves TOWARDS the eye

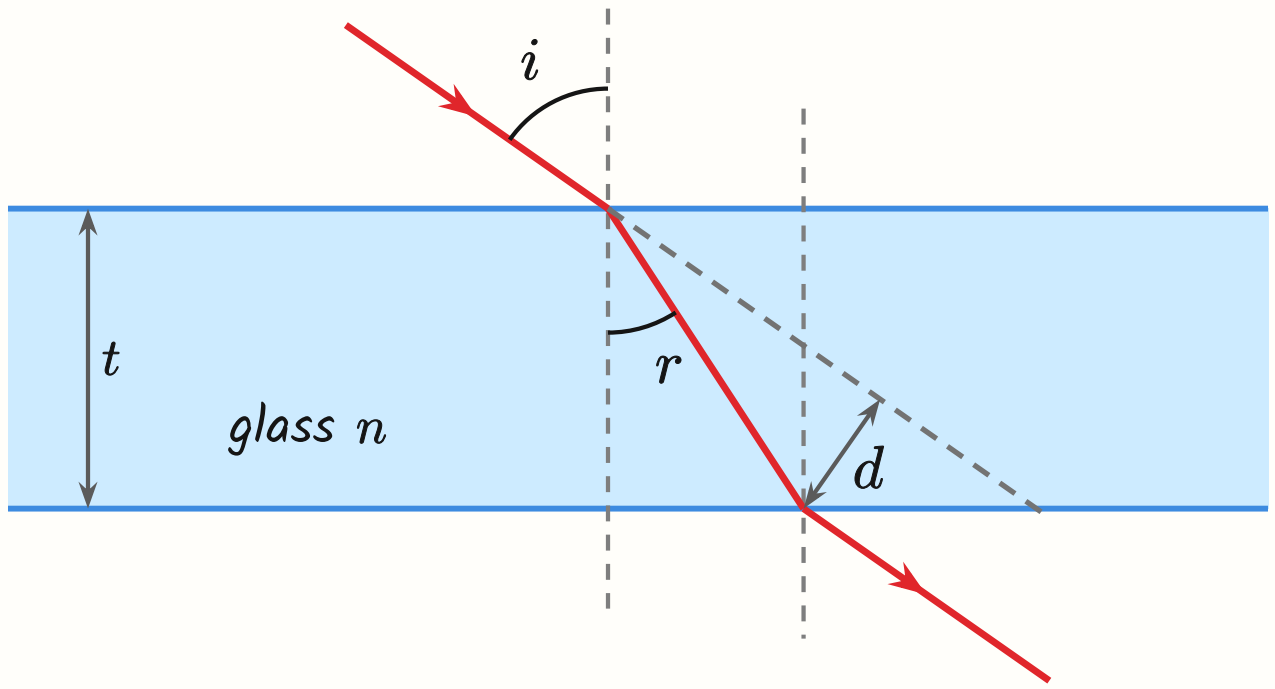
- Object in AIR seen from inside water (fish sees a bird): it looks FARTHER, $h' = nh$ (near-normal).

Example

$6\,\text{cm}$ glass (1.5) on $8\,\text{cm}$ water $(4/3)$:
$h' = \frac{6}{1.5} + \frac{8}{4/3} = 4 + 6 = 10\,\text{cm}$ (real $14\,\text{cm}$). Slab $t = 6\,\text{cm}$, $n = 1.5$: shift $2\,\text{cm}$.

Lateral Shift through a Slab

1 Parallel glass slab



$$d = \frac{t\,\sin(i - r)}{\cos r}$$

PARALLEL-faced slab, SAME medium on both sides · emergent ray ∥ incident · d: perpendicular gap

2 Limits

- $i = 0$: $d = 0$. *Grazing* ($i \to 90°$): $d \to t$.
- *Small* i *(radians)*: $d \approx t\,i\left(1 - \frac{1}{n}\right)$.

Example

$t = 3\,\text{cm}$, $n = \sqrt{3}$, $i = 60°$: $\sin r = \frac{\sin 60°}{\sqrt{3}} = \frac{1}{2}$, $r = 30°$;

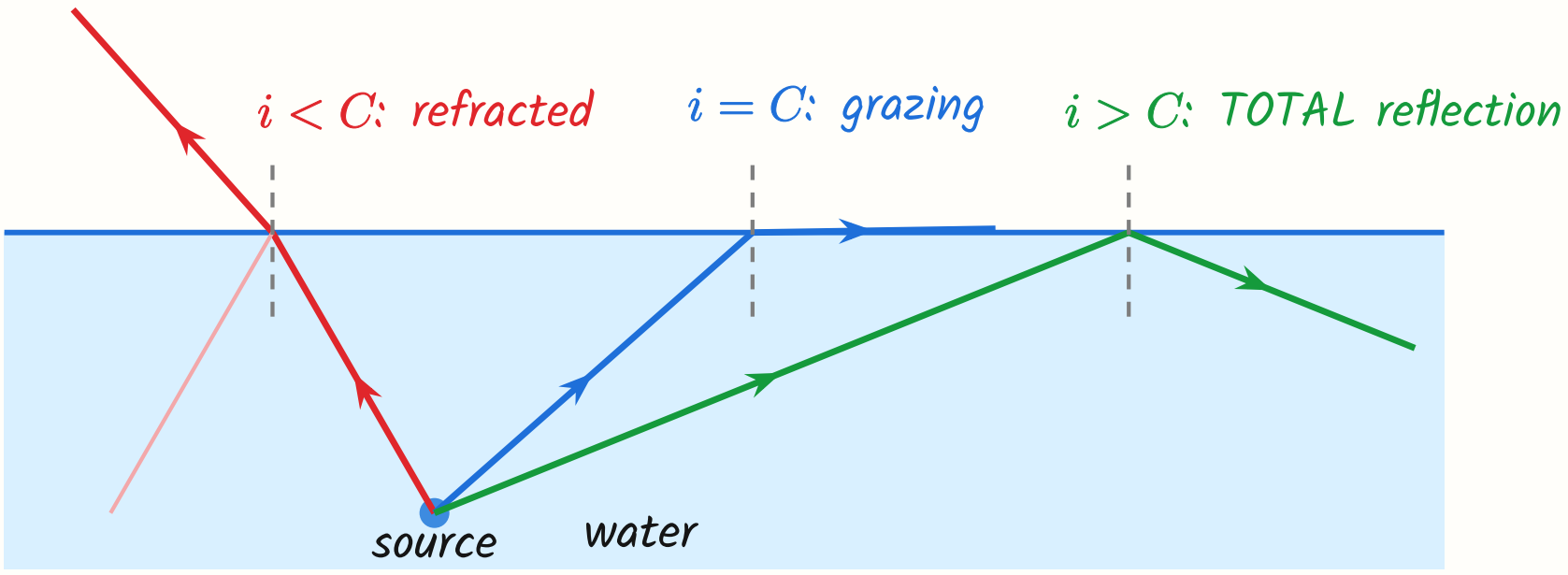
$d = \frac{3\sin 30°}{\cos 30°} = \sqrt{3} \approx 1.73\,\text{cm}$.

Golden points

★ *LATERAL shift (oblique ray, sideways) is not the NORMAL shift* $t(1 - 1/n)$ *(image seen through a slab, near-normal).*

Total Internal Reflection

1 From denser to rarer



$$\sin C = \frac{n_{rarer}}{n_{denser}} \quad \left(= \frac{1}{n} \text{ into air}\right)$$

TIR only when light goes from DENSER to RARER and $i > C$ · 100 % reflected (no loss, unlike a metal mirror)

Example

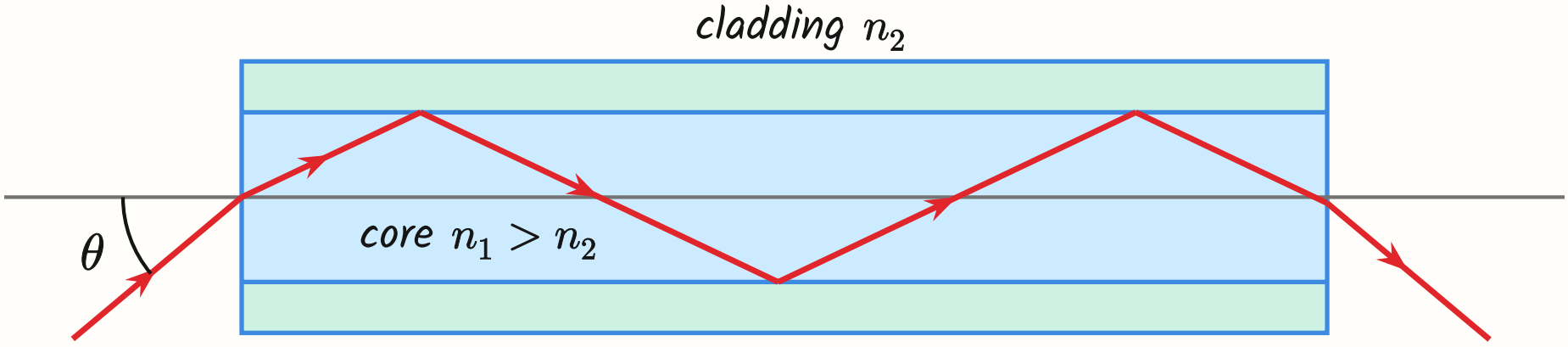
Critical angle: water $(4/3)$ $48.6°$; glass (1.5) $41.8°$; diamond (2.42) $24.4°$.

2 Uses

- Diamond sparkles: small C, cut so that light totally reflects many times.
- Mirage: hot air near the road has smaller n; light from the sky bends up and is totally reflected at grazing angles.
- Totally reflecting prism $(45°–90°–45°)$ turns rays by $90°$ or $180°$; needs $C < 45°$, i.e. $n > \sqrt{2}$ (periscopes, binoculars).

Optical Fibre and Circle of Light

1 Optical fibre



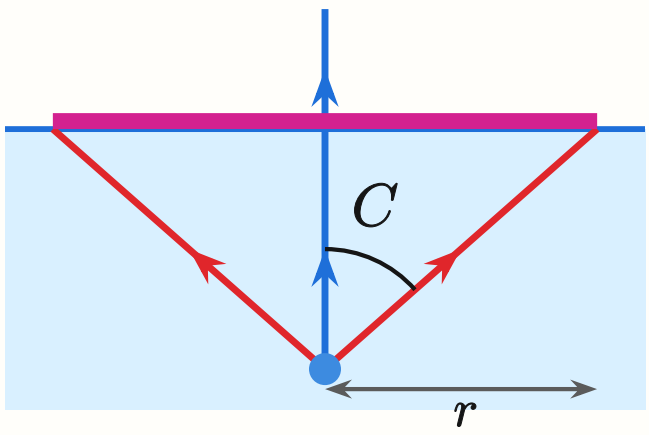
$$\sin\theta_a = \sqrt{n_1^2 - n_2^2} \quad \text{(numerical aperture)}$$

ray enters the end face from AIR; TIR at the core–cladding wall needs $\theta \le \theta_a$ · from a medium n_0: $n_0 \sin\theta_a = \sqrt{n_1^2 - n_2^2}$ · if $\sqrt{n_1^2 - n_2^2} \ge n_0$ every ray is guided

Example

$n_1 = 1.5$, $n_2 = 1.2$: $\sqrt{2.25 - 1.44} = 0.9$, $\theta_a = 64.2°$ (drawn ray: $\theta = 40°$).

2 Circle of light



$$r = h\tan C = \frac{h}{\sqrt{n^2 - 1}}$$

point source at depth h in a liquid of index n under air: light leaves only through this circle

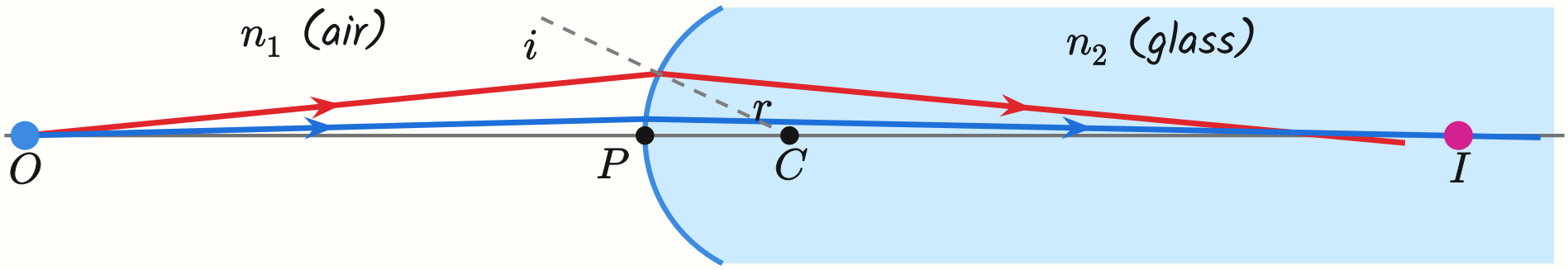
Example

Water, $h = 4\,\text{m}$:

$$r = \frac{4}{\sqrt{16/9 - 1}} = \frac{12}{\sqrt{7}} \approx 4.54\,\text{m}.$$

Refraction at a Spherical Surface

1 Single curved surface



$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}, \qquad m = \frac{n_1 v}{n_2 u}$$

PARAXIAL rays (rays traced exactly: the near-axis blue ray crosses almost at I, the wide red ray falls short) · Cartesian from the vertex P · light from n_1 into n_2 · $R > 0$ if C is on the outgoing side

Example

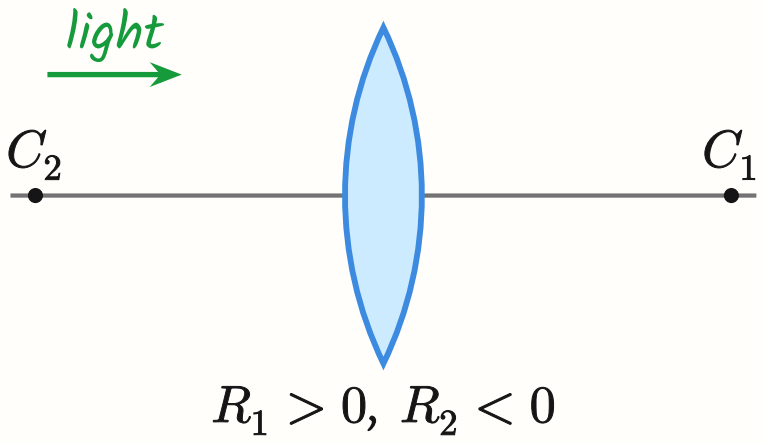
Object in air $40\,\text{cm}$ from convex glass (1.5), $R = 10\,\text{cm}$: $\frac{1.5}{v} = \frac{0.5}{10} - \frac{1}{40}$, $v = 60\,\text{cm}$; $m = \frac{60}{1.5\times(-40)} = -1$.

2 Special cases

- *Plane surface ($R \to \infty$):* $v = \frac{n_2}{n_1}u$, *the apparent-depth result (p9).*
- *Power of the surface* $P = \frac{n_2 - n_1}{R}$; *focal lengths* $f_1 = -\frac{n_1 R}{n_2 - n_1}$, $f_2 = \frac{n_2 R}{n_2 - n_1}$.
- *Also for concave surfaces ($R < 0$) and glass to air (n_1: medium of the INCIDENT light).*

Thin Lens: Lensmaker's Formula

1 Lensmaker's formula



$$\frac{1}{f} = (n_{rel} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$n_{rel} = n_{lens}/n_{medium}$ · THIN lens, paraxial, SAME medium both sides · R_1: surface met first, $R > 0$ if centre on the outgoing side

- Equi-biconvex: $f = \frac{R}{2(n-1)}$ $(= R$ for $n = 1.5)$. Plano-convex: $f = \frac{R}{n-1}$ $(= 2R)$.

2 Lens in a liquid

$$\frac{f_{liquid}}{f_{air}} = \frac{n - 1}{n/n_l - 1}$$

same thin lens (index n) in air and in a liquid n_l

- Glass (1.5) in water $(4/3)$: $\frac{0.5}{1.125-1} = 4$, so $f_w = 4f_a$.
- $n_l > n$: convex lens DIVERGES (air bubble in water); $n_l = n$: $f \to \infty$.

3 Different media on the two sides

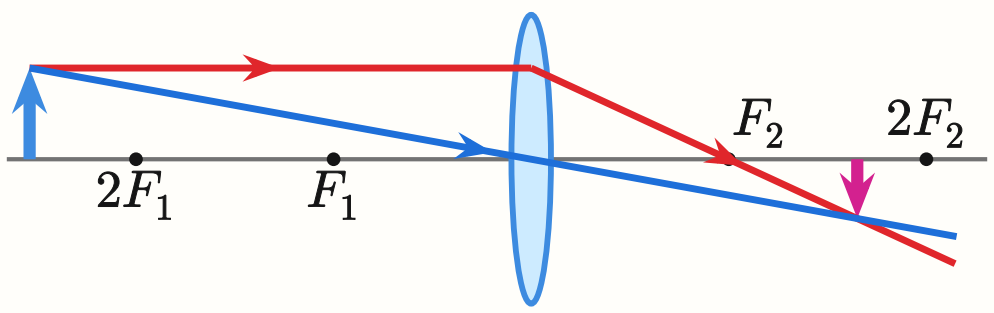
$$\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2}$$

thin lens n_2, object side n_1, image side n_3, paraxial · $f_2/f_1 = -n_3/n_1$

Lens Ray Diagrams

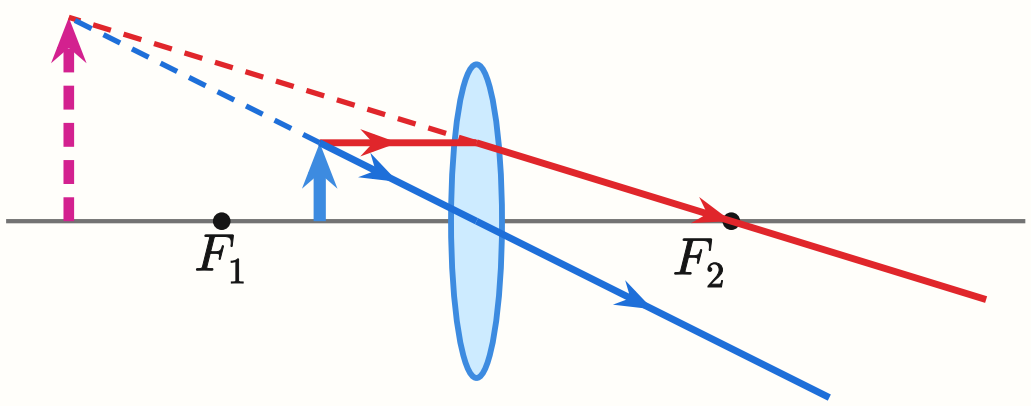
thin lenses, paraxial rays; heights exaggerated for clarity

1 Convex lens, object beyond $2F$



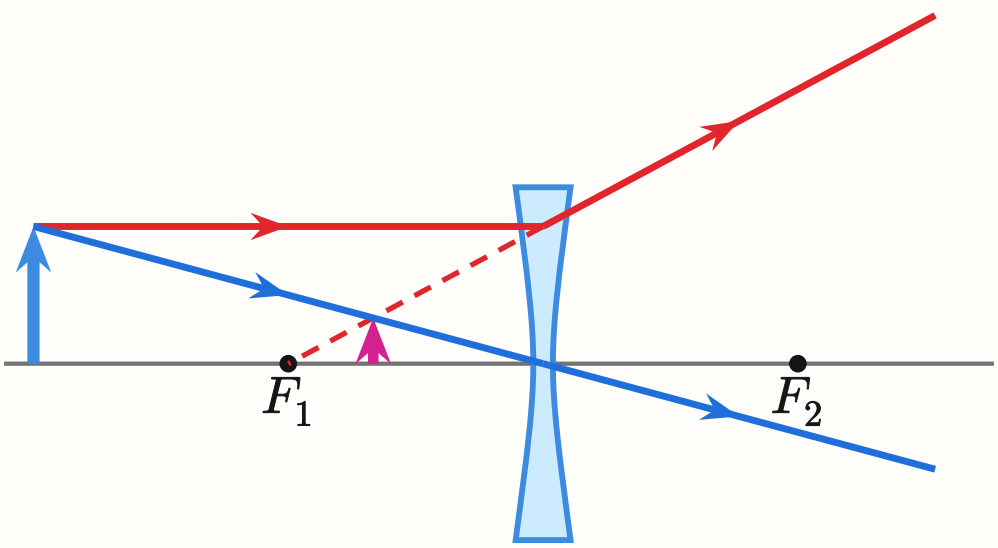
- Image between F_2 and $2F_2$: REAL, inverted, diminished (camera).
- Red: parallel → through F_2. Blue: through the optical centre, undeviated.

2 Convex lens, object inside F



- Image on the SAME side: VIRTUAL, erect, magnified (magnifying glass).
- Dashed: backward extensions meet at the image.

3 Concave lens, any real object



- Image between F_1 and the lens: VIRTUAL, erect, DIMINISHED.
- Red: parallel → diverges as if from F_1.

Lens Formula

1 Lens formula and magnification

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f}, \qquad m=\frac{h'}{h}=\frac{v}{u}$$

THIN lens, paraxial, same medium both sides · Cartesian, from the optical centre, real object $u<0$ · convex $f>0$, concave $f<0$

2 Where is the image? (convex lens)

Object	Image	Nature
at ∞	at F_2	real, inverted, point size
beyond $2F_1$	F_2 to $2F_2$	real, inverted, diminished
at $2F_1$	at $2F_2$	real, inverted, same size
F_1 to $2F_1$	beyond $2F_2$	real, inverted, magnified
at F_1	at ∞	real, inverted, very large
F_1 to O	object side	VIRTUAL, erect, magnified
concave: any	F_1 to O	VIRTUAL, erect, diminished

3 Two standard results

$$D_{min}=4f \qquad v_I=m^2 v_O$$

$4f$: least real-object to REAL-image distance (convex lens, $u=-2f$) · m^2: object moving ALONG the axis, lens at rest, image moves the SAME way

Example

$f=20\,\text{cm}$, *object at* $30\,\text{cm}$: $v=60\,\text{cm}$, $m=-2$; *object approaching at* $1\,\text{cm}\,\text{s}^{-1}$: *image moves AWAY at* $4\,\text{cm}\,\text{s}^{-1}$.

Power and Lens Combinations

1 Power of a lens

$$P = \frac{1}{f\,(\text{in } m)} \qquad \text{dioptre: } 1\,\mathrm{D} = 1\,\mathrm{m}^{-1}$$

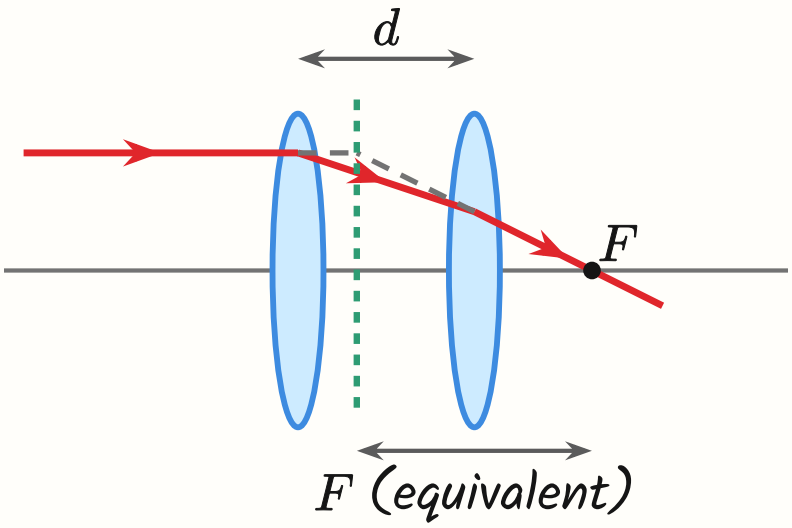
lens in AIR · convex (converging) $P > 0$, concave (diverging) $P < 0$

2 Thin lenses in contact

$$P = P_1 + P_2 + \cdots, \qquad m = m_1 m_2 \cdots$$

THIN lenses in CONTACT, coaxial · $1/F = 1/f_1 + 1/f_2 + \cdots$

3 Separated by a distance d



$$P = P_1 + P_2 - d\,P_1 P_2$$

two coaxial THIN lenses in air · F is measured from the EQUIVALENT lens (green), not from either lens

Example

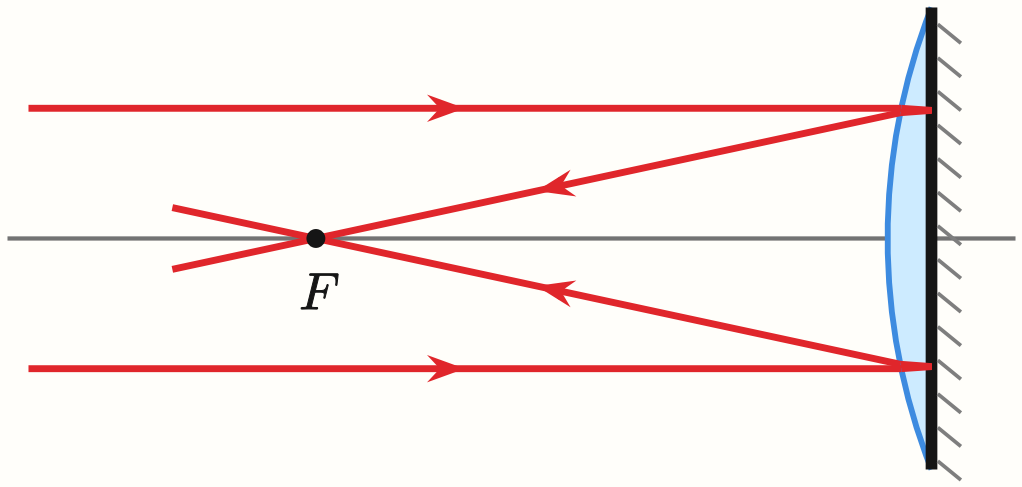
$f_1 = f_2 = 10\,\mathrm{cm}$, $d = 5\,\mathrm{cm}$:
$\frac{1}{F} = \frac{1}{10} + \frac{1}{10} - \frac{5}{100} = 0.15$,
$F = 6.67\,\mathrm{cm}$.

4 Cutting a lens

- Cut ALONG the axis: each half keeps f, full image, less light.
- Equi-biconvex cut PERPENDICULAR to the axis: two plano-convex lenses of $2f$ each.

Silvered Lens and Lens–Mirror Systems

1 Silvered lens = a mirror



plano-convex lens, PLANE face silvered (rays traced exactly)

$$P_{eq} = 2P_L + P_M, \qquad f_{eq} = -\frac{1}{P_{eq}}$$

thin lens silvered on one face, paraxial: light crosses the lens TWICE · $P_L = 1/f_L$, $P_M = -1/f_M$ *(concave* $P_M > 0$*) ·* $P_{eq} > 0$*: CONCAVE mirror*

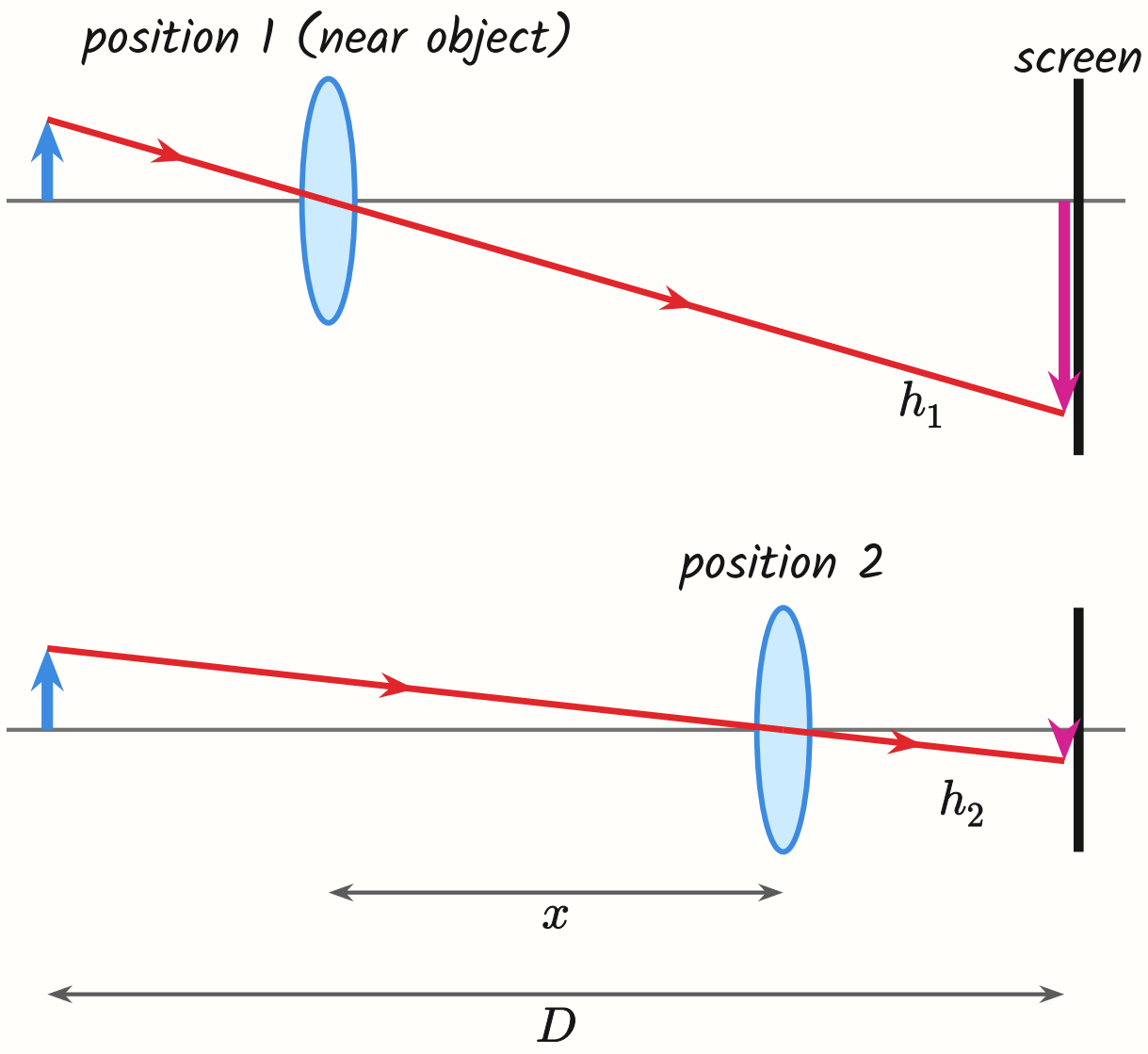
- *Plano-convex: PLANE face silvered* $f = \frac{R}{2(n-1)}$*; CURVED face silvered* $f = \frac{R}{2n}$*.* ($n = 1.5$, $R = 20\,\text{cm}$: $20\,\text{cm}$ *and* $\frac{20}{3} \approx 6.67\,\text{cm}$.)

2 Lens and mirror apart

- *Image by the lens* $\rightarrow$ *object for the mirror (may be virtual)* $\rightarrow$ *image by the mirror* $\rightarrow$ *back through the lens with light going RIGHT to LEFT (re-apply the convention from that side).* $m = m_1 m_2 m_3$.
- *Object at* F *of a convex lens, plane mirror behind: light returns onto the object (finds* f*).*

Displacement Method

1 Two lens positions, one screen



$$f = \frac{D^2 - x^2}{4D}, \qquad h = \sqrt{h_1 h_2}, \qquad m_1 m_2 = 1$$

convex lens, object–screen distance D FIXED and $D > 4f$ · x: distance between the two lens positions (u and v swap) · $D = 4f$: only one position

Example

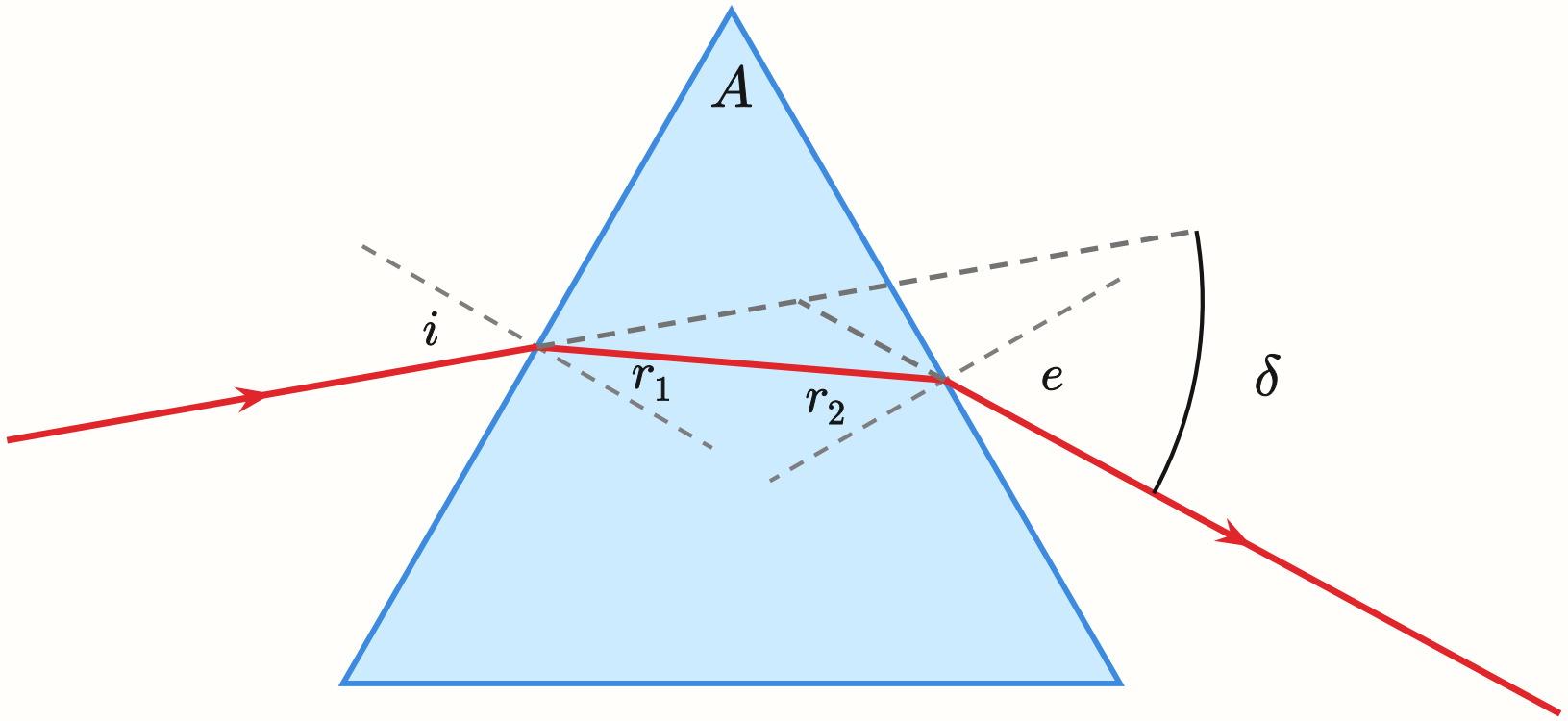
$D = 90\,\text{cm}$, $x = 30\,\text{cm}$: $f = \frac{8100-900}{360} = 20\,\text{cm}$. *Images* $8\,\text{cm}$ *and* $2\,\text{cm}$: *object* $\sqrt{16} = 4\,\text{cm}$.

Golden points

★ $D < 4f$: *no real image on the screen for any lens position. Both images are real and inverted,* $h_1 h_2 = h^2$.

Prism: Deviation

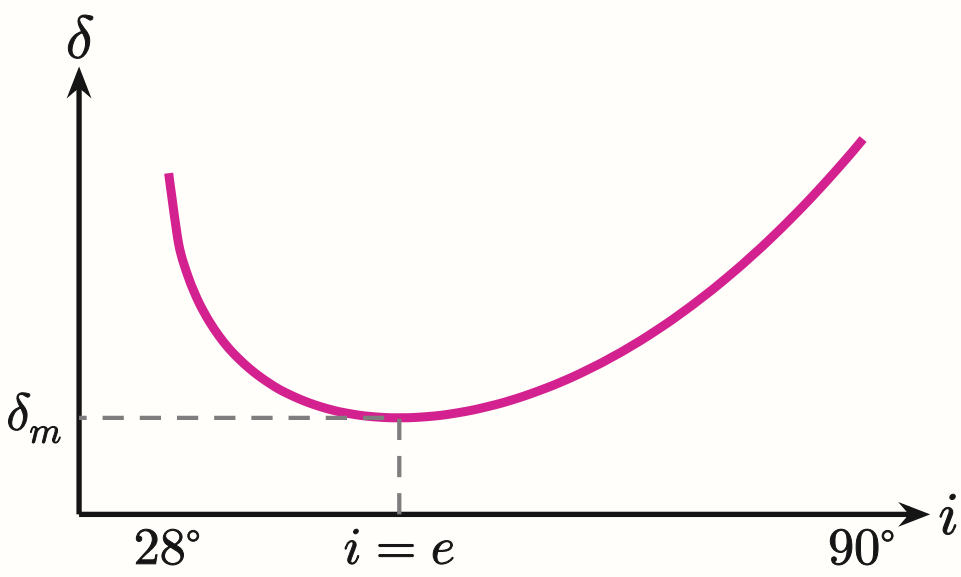
1 Path through a prism



$$r_1 + r_2 = A, \qquad \delta = i + e - A$$

prism of angle A in air, ray in a principal section · drawn exactly: $A = 60°$, $n = 1.5$, $i = 40°$ gives $e = 58.5°$, $\delta = 38.5°$

2 Deviation against incidence



- *Drawn for $A = 60°$, $n = 1.5$: one MINIMUM $\delta_m = 37.2°$ at $i = e = 48.6°$.*
- *Same δ for i and e interchanged (reversibility), so the graph is not symmetric in i but each $\delta > \delta_m$ occurs twice.*

Minimum Deviation and Thin Prism

1 Minimum deviation

$$n = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

prism in air · at δ_m: $i = e$, $r_1 = r_2 = A/2$, ray inside ∥ base (isosceles prism)

Example

Equilateral, $n = 1.5$: $\sin\frac{60°+\delta_m}{2} = 0.75$, so $\frac{60°+\delta_m}{2} = 48.6°$, $\delta_m = 37.2°$.

2 Thin prism

$$\delta = (n-1)A$$

SMALL A AND small (near-normal) i · then independent of i · $A = 6°$, $n = 1.5$: $\delta = 3°$

3 Limiting cases

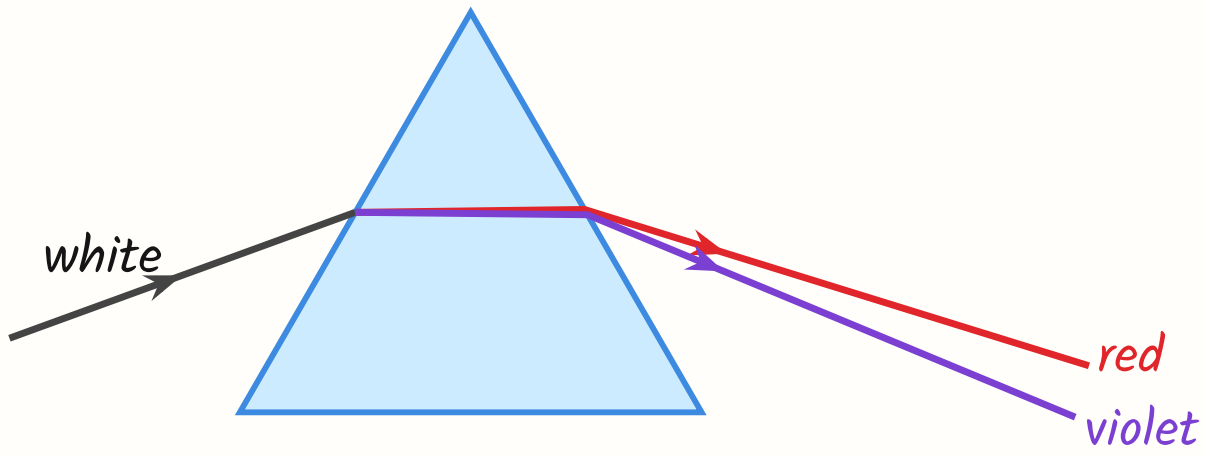
- *Grazing incidence ($i = 90°$): $r_1 = C$, deviation MAXIMUM (same at grazing emergence).*
- *Emerges only if $r_2 \le C$: $i \ge i_{min}$, $\sin i_{min} = n \sin(A - C)$ ($A > C$).*
- *NO ray emerges for any i if $A > 2C$.*

Example

$A = 60°$, $n = 1.5$ ($C = 41.8°$): $i_{min} = 27.9°$; $\delta_{max} = 90° + 27.9° - 60° = 57.9°$.

Dispersion and Scattering

1 Dispersion by a prism



traced exactly, $n_r = 1.50$, $n_v = 1.56$ *(spread exaggerated)*

$$\theta = \delta_v - \delta_r = (n_v - n_r)A, \qquad \omega = \frac{n_v - n_r}{n_y - 1} = \frac{\theta}{\delta_y}$$

THIN prism, small i · θ*: angular dispersion* · ω*: dispersive power (material only, not* A*)* · $\delta_y = (n_y - 1)A$*: mean deviation*

- n *depends on colour (Cauchy:* $n = a + b/\lambda^2$*), larger for violet: violet deviates MOST, red LEAST.*

Example

Crown: $n_r = 1.513$, $n_y = 1.517$, $n_v = 1.523$, $A = 10°$: $\theta = 0.1°$, $\delta_y = 5.17°$, $\omega = \frac{0.010}{0.517} = 0.0193$.

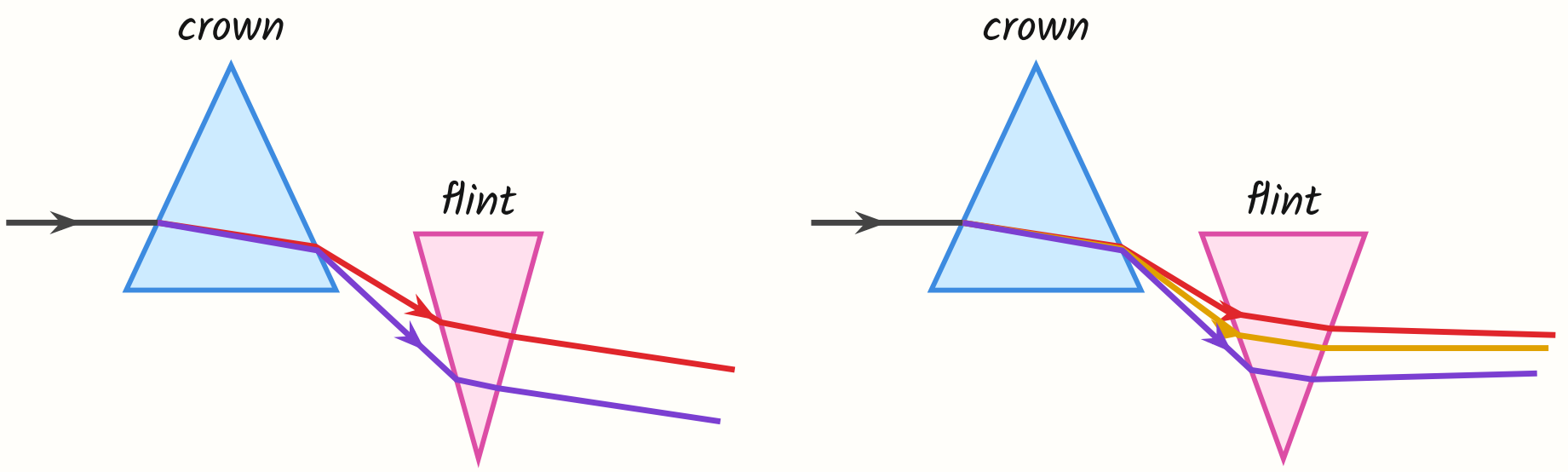
2 Scattering (basics)

$$I_{scattered} \propto \frac{1}{\lambda^4}$$

Rayleigh: particles much SMALLER than λ *(air molecules)* · *blue sky; red Sun at sunrise and sunset (long path scatters the blue away)* · *white clouds: droplets* $\gg \lambda$ *scatter all colours almost equally*

Achromatic and Direct Vision

1 Two prisms, one inverted



ACHROMATIC: red ‖ violet — DIRECT VISION: yellow undeviated

exact trace, n exaggerated (r/y/v): crown $1.50/1.56/1.62$, flint $1.62/1.74/1.86$

2 Achromatic: deviation without dispersion

$$(n_v - n_r)A + (n_v' - n_r')A' = 0 \;\Leftrightarrow\; \omega\delta + \omega'\delta' = 0$$

THIN prisms, the second INVERTED (A', δ' negative) · net $\delta + \delta' = \delta\left(1 - \frac{\omega}{\omega'}\right)$ · lens doublet in contact: $\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$

3 Direct vision: dispersion without deviation

$$(n_y - 1)A + (n_y' - 1)A' = 0$$

thin prisms, second inverted · mean (yellow) ray undeviated, net dispersion $\delta(\omega - \omega') \neq 0$

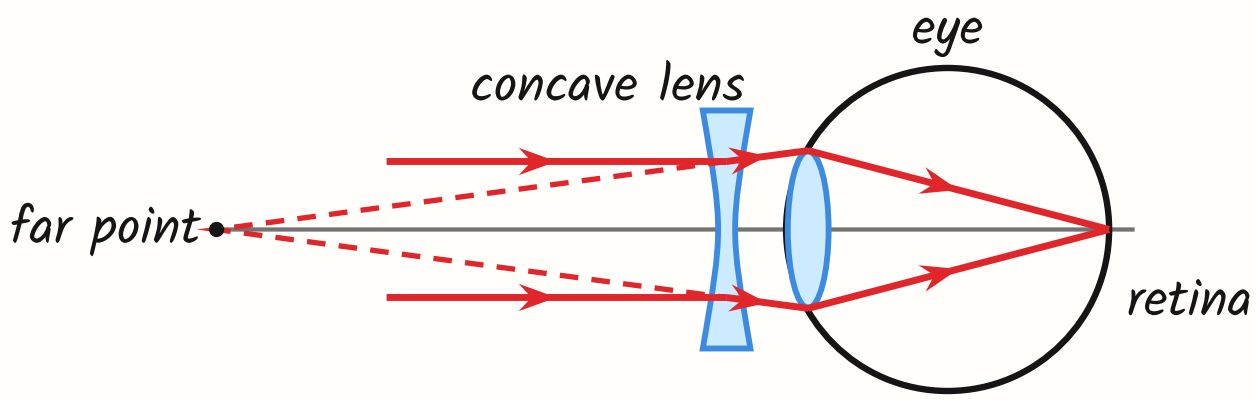
Example

Crown $A = 6°$, $n_y = 1.5$, $\omega = 0.02$ ($\delta = 3°$); flint $n_y' = 1.6$, $\omega' = 0.03$. Achromatic: $\delta' = -2°$, $A' = 3.33°$, net $1°$. Direct vision: $A' = 5°$, net dispersion $-0.03°$.

Human Eye and its Defects

normal eye: near point $25\,\mathrm{cm}$, far point ∞; lenses close to the eye

1 Myopia (short sight)



- *Far point at FINITE x: a distant object focuses IN FRONT of the retina.*
- *CONCAVE lens $f = -x$ $(P = -1/x)$: parallel rays seem to come from the far point. Far point $2\,\mathrm{m}$: $P = -0.5\,\mathrm{D}$.*

2 Hypermetropia (long sight)

$$\frac{1}{f} = \frac{1}{25} - \frac{1}{y} \quad (cm)$$

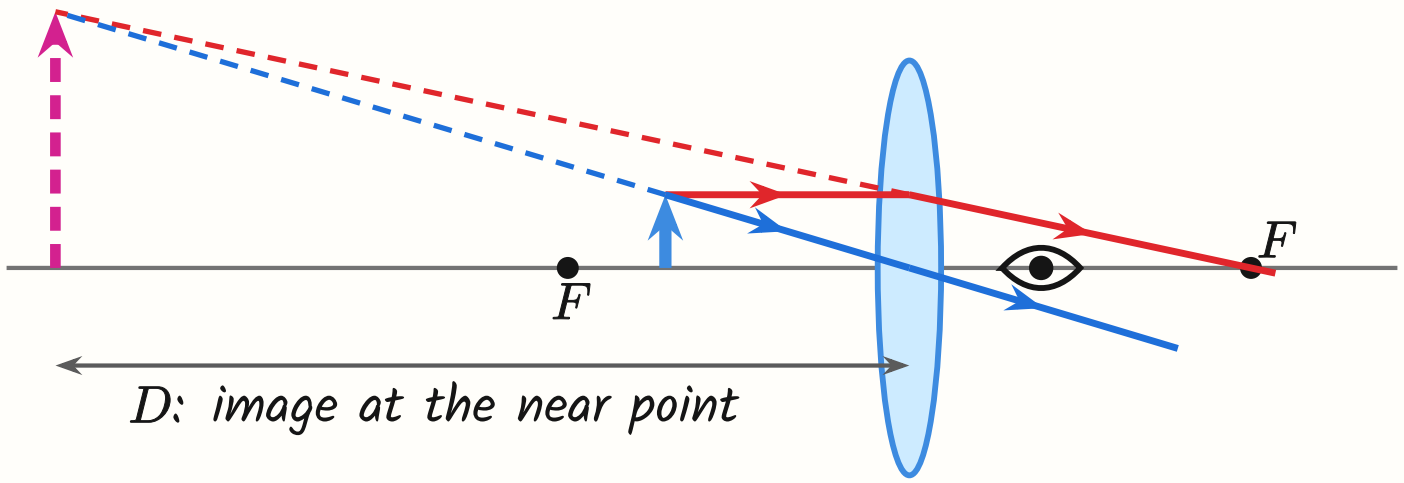
near point at $y > 25\,\mathrm{cm}$ · CONVEX lens: object at $25\,\mathrm{cm}$ appears at y · $y = 50\,\mathrm{cm}$: $f = +50\,\mathrm{cm}$, $P = +2\,\mathrm{D}$

3 Accommodation, presbyopia, astigmatism

- *Ciliary muscles change the eye-lens f; power of accommodation $\frac{1}{d_{near}} - \frac{1}{d_{far}}$ $(m) \approx 4\,\mathrm{D}$ (normal young eye).*
- *Presbyopia (age): bifocals. Astigmatism (cornea curved unequally): CYLINDRICAL lens.*

Simple Microscope

1 Magnifying glass



$$M = \frac{\beta}{\alpha} = 1 + \frac{D}{f} \quad \text{(image at } D\text{)}$$

$$M = \frac{D}{f} \quad \text{(image at } \infty\text{, relaxed eye)}$$

thin convex lens held CLOSE to the eye, small angles · $\alpha = h/D$: angle of the object at the near point, naked eye · β: angle of the image · $D = 25\,\text{cm}$

2 Remember

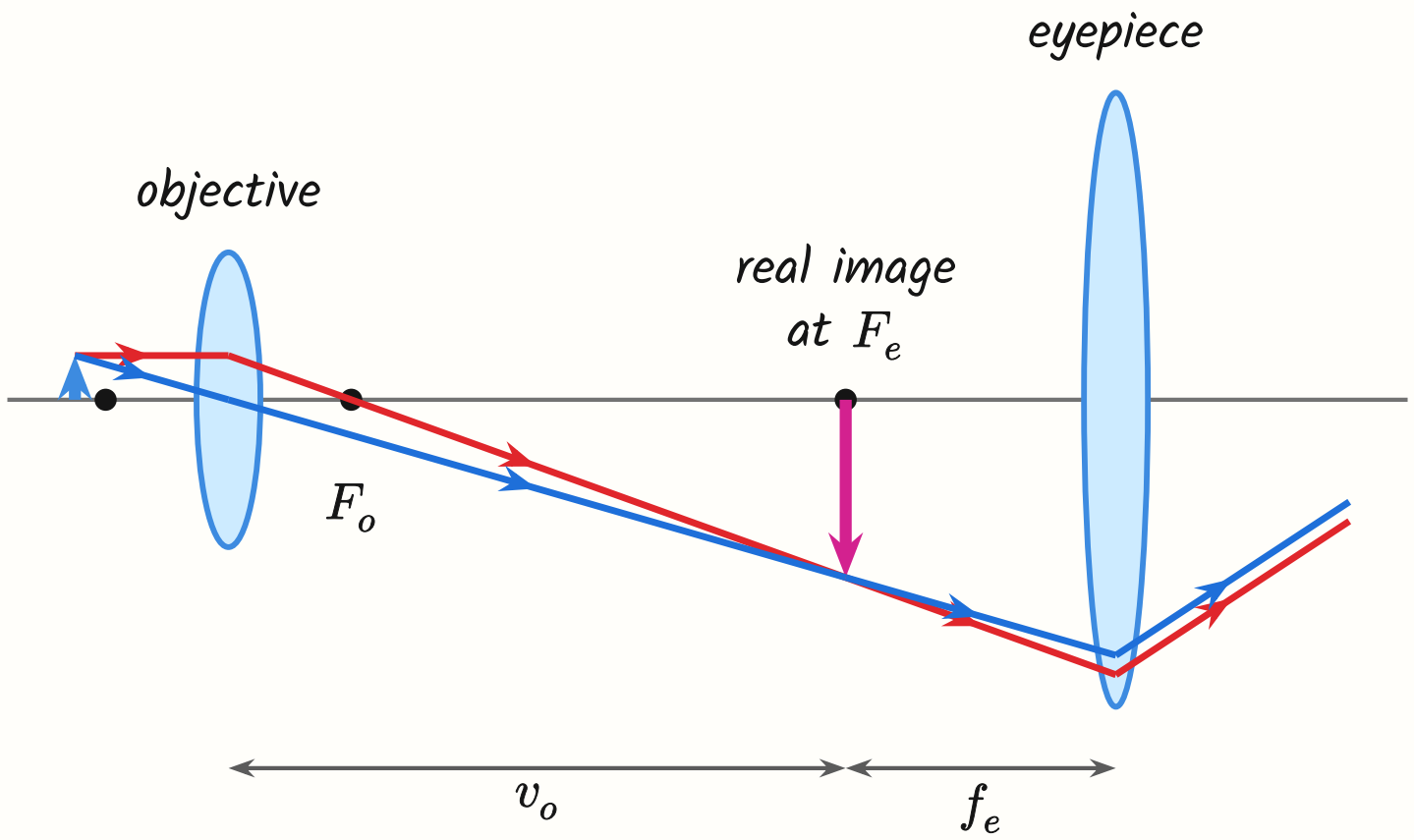
- Image at D: object just inside F, M is largest; image at ∞: object AT F, eye relaxed (normal use).
- M is ANGULAR magnification; only for the image at D does it equal the linear $m = v/u = 1 + D/f$.
- Useful only if $f < D$.

Example

$f = 5\,\text{cm}$: $M = 1 + \frac{25}{5} = 6$ (image at D); $M = \frac{25}{5} = 5$ (relaxed eye).

Compound Microscope

1 Ray diagram (normal adjustment)



$$|M| = \frac{v_o}{|u_o|} \cdot \frac{D}{f_e} \quad \text{(final image at } \infty\text{)}$$

$$|M| = \frac{v_o}{|u_o|}\left(1 + \frac{D}{f_e}\right) \quad \text{(final image at } D\text{)}$$

thin lenses, paraxial, eye close to the eyepiece · final image inverted and virtual · length $v_o + f_e$ (at ∞) or $v_o + |u_e|$ (at D)

- *Object just outside F_o, intermediate image near F_e:* $|M| \approx \dfrac{L}{f_o} \cdot \dfrac{D}{f_e}$

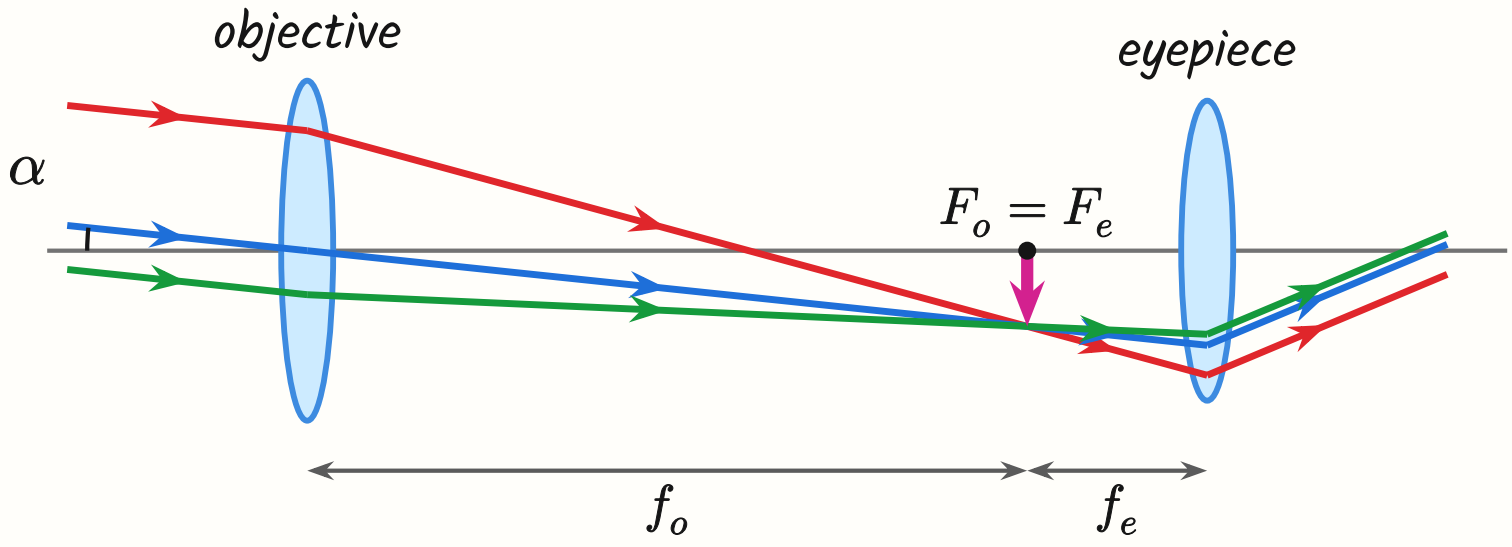
  (L: tube length, from F_o' to F_e). Small f_o and f_e give large M.

Example

$f_o = 2\,\text{cm}$, $u_o = -2.5\,\text{cm}$: $v_o = 10\,\text{cm}$, $m_o = -4$; $f_e = 5\,\text{cm}$,
relaxed eye: $|M| = 4 \times \frac{25}{5} = 20$, *length* $15\,\text{cm}$.

Astronomical Telescope

1 Ray diagram (normal adjustment)



$$M = \frac{\beta}{\alpha} = \frac{f_o}{f_e}, \qquad L = f_o + f_e$$

NORMAL adjustment: object and final image at infinity · thin lenses, small angles · image inverted

$$M = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right), \qquad L = f_o + \frac{f_e D}{f_e + D}$$

final image at the near point $D = 25\,\text{cm}$

Example

$f_o = 150\,\text{cm}$, $f_e = 5\,\text{cm}$: normal $M = 30$, $L = 155\,\text{cm}$; image at D: $M = 30\left(1 + \frac{5}{25}\right) = 36$.

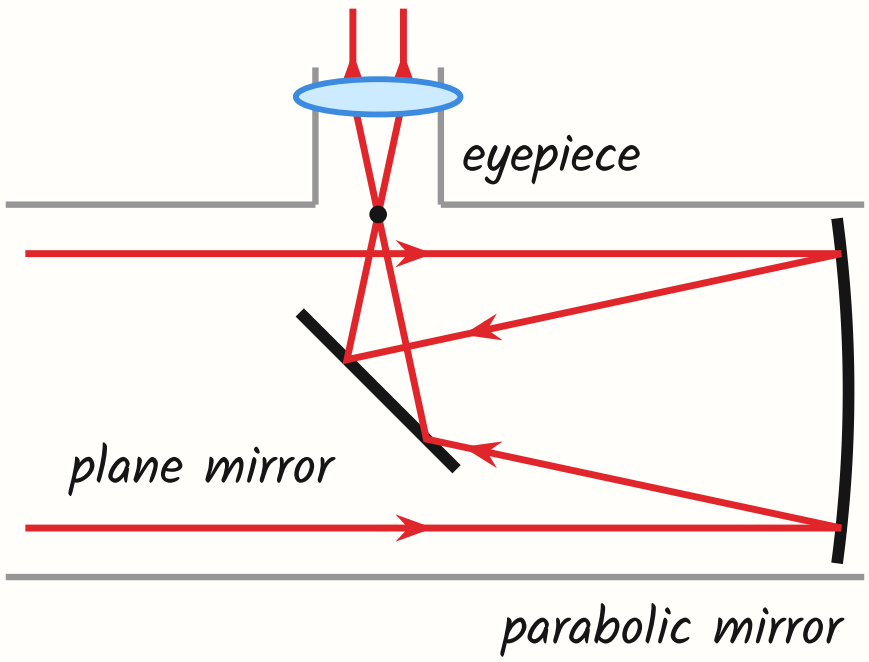
2 Upright images (normal adjustment)

- Terrestrial: erecting lens (f) in between: $L = f_o + f_e + 4f$, $M = f_o/f_e$, erect.
- Galilean: CONCAVE eyepiece: $L = f_o - |f_e|$, $M = f_o/|f_e|$, erect.

Reflecting Telescopes

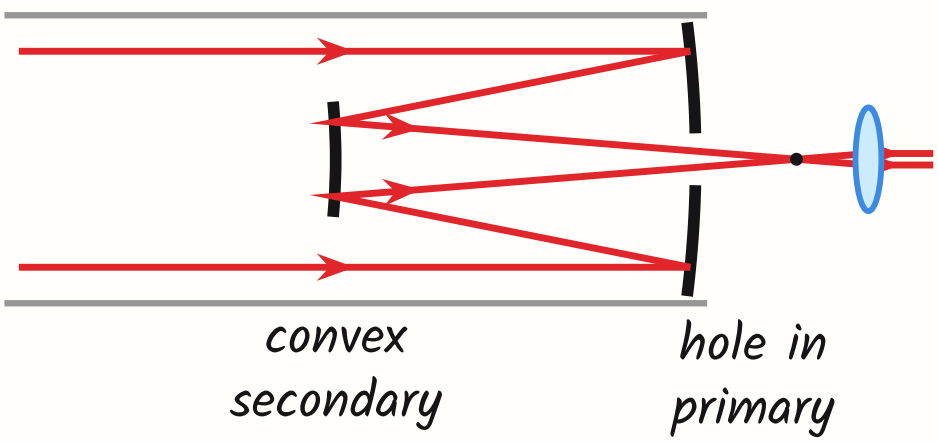
rays traced exactly; parabolic primary mirrors

1 Newtonian telescope



- Parabolic primary sends ALL rays parallel to its axis to its focus (no spherical aberration).
- A plane mirror at $45°$ turns the beam sideways to the eyepiece.

2 Cassegrain telescope



- Convex (hyperbolic) secondary sends the light back through a HOLE in the primary: short tube for a long focal length.

3 Why mirrors?

- No chromatic aberration; parabolic: no spherical aberration.
- Big mirror supported from the back, cheaper than a lens: large aperture, more light, better resolution.
- $M = f_o/f_e$, $f_o = R/2$ of the primary (normal adjustment, paraxial).

Common Traps

1 Exam traps

Avoid these

- ✗ $f = R/2$ for every ray (only PARAXIAL rays; wide rays focus closer).
- ✗ A mirror's f changes in water (no: $f = R/2$, independent of the medium; a LENS's f does change).
- ✗ Apparent depth h/n at any viewing angle (only NEAR-NORMAL viewing).
- ✗ TIR from air into glass (only DENSER to RARER, and only for $i > C$).
- ✗ Lateral shift $= t(1 - 1/n)$ (that is the NORMAL shift; lateral shift is $t \sin(i - r)/\cos r$ for a parallel slab).
- ✗ $\delta = (n - 1)A$ for any prism (only THIN prism, small i).
- ✗ Telescope $M = f_o/f_e$ always (only NORMAL adjustment; image at D: $\frac{f_o}{f_e}(1 + \frac{f_e}{D})$).
- ✗ Convex lens always converges (only if $n_{lens} > n_{medium}$).
- ✗ $v_I = -m^2 v_O$ for a lens too (lens: $v_I = +m^2 v_O$; both along the axis).
- ✗ Taking $u > 0$ for a real object (Cartesian, light from the left: real object $u < 0$).
- ✗ Dispersive power depends on A (no: material only, thin prism).

Golden points

- ★ First ask: paraxial? thin lens? which medium? final image at ∞ or at D?

Standard Results

1 One-line results

Quantity	Result	Condition
Mirror	$\frac{1}{v}+\frac{1}{u}=\frac{1}{f},\ m=-\frac{v}{u}$	paraxial, Cartesian
Lens	$\frac{1}{v}-\frac{1}{u}=\frac{1}{f},\ m=\frac{v}{u}$	thin, paraxial
Lensmaker	$\frac{1}{f}=(n-1)(\frac{1}{R_1}-\frac{1}{R_2})$	thin lens in air
Apparent depth	h/n	near-normal viewing
Lateral shift	$t\sin(i-r)/\cos r$	parallel slab
Critical angle	$\sin C=1/n$	denser into air
Prism	$n=\frac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}}$	minimum deviation
Thin prism	$\delta=(n-1)A$	small A, small i
Microscope	$\frac{L}{f_o}\cdot\frac{D}{f_e}$	approx., image at ∞
Telescope	$f_o/f_e,\ L=f_o+f_e$	normal adjustment

2 Numbers to remember

- Critical angle: glass (1.5) $41.8°$, water $(4/3)$ $48.6°$. Thin glass (1.5) lens in water: $f_w=4f_a$.
- Silvered plano-convex: $R/2(n-1)$ (plane face) or $R/2n$ (curved face). Equilateral glass prism (1.5): $\delta_m=37.2°$.

Summary

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$ • $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ • $n_1\sin i=n_2\sin r$ • $P=P_1+P_2$

mirror, thin lens: paraxial, Cartesian · P_1+P_2: thin lenses in contact

Questions: JEE Main Level (I)

paraxial rays, thin lenses in air, Cartesian sign convention

Q1 · JEE Main level · MCQ

Object $15\,\text{cm}$ in front of a concave mirror, $|f| = 10\,\text{cm}$. Image: (A) real, inverted, $30\,\text{cm}$ in front, $2\times$ (B) virtual, erect, $30\,\text{cm}$ behind (C) real, inverted, $6\,\text{cm}$ in front (D) virtual, erect, $6\,\text{cm}$ behind

Solution: $u = -15$, $f = -10$: $\frac{1}{v} = -\frac{1}{10} + \frac{1}{15} = -\frac{1}{30}$, $v = -30\,\text{cm}$ (real, in front); $m = -\frac{v}{u} = -2$.

Answer: (A)

Q2 · JEE Main level · Numerical

A coin lies under $20\,\text{cm}$ of water ($n = 4/3$). Seen from directly above (near-normal), by how many cm does it appear raised?

Solution: $h' = \frac{20}{4/3} = 15\,\text{cm}$; rise $= 20 - 15 = 5\,\text{cm}$.

Answer: 5

Q3 · JEE Main level · MCQ

Light travels at $2 \times 10^8\,\text{m s}^{-1}$ in a medium. Critical angle, medium to air: (A) $\sin^{-1}(2/3)$ (B) $\sin^{-1}(1/3)$ (C) $\sin^{-1}(3/4)$ (D) $\sin^{-1}(1/2)$

Solution: $n = \frac{3\times10^8}{2\times10^8} = 1.5$; $\sin C = \frac{1}{n} = \frac{2}{3}$, $C \approx 41.8°$.

Answer: (A)

Q4 · JEE Main level · Numerical

A thin equi-biconvex glass lens ($n = 1.5$, $|R| = 20\,\text{cm}$) has $f = 20\,\text{cm}$ in air. Its focal length (in cm) in water ($n = 4/3$)?

Solution: $\frac{1}{f_w} = \left(\frac{1.5}{4/3} - 1\right)\left(\frac{1}{20} + \frac{1}{20}\right) = \frac{1}{8} \cdot \frac{1}{10} = \frac{1}{80}$.

Answer: 80

Questions: JEE Main Level (2)

paraxial rays, thin lenses in air, $D = 25\,\text{cm}$

Q5 · JEE Main level · MCQ

Thin lenses of $+20\,\text{cm}$ and $-30\,\text{cm}$ in contact. The pair is: (A) converging, $60\,\text{cm}$ (B) diverging, $60\,\text{cm}$ (C) converging, $12\,\text{cm}$ (D) diverging, $12\,\text{cm}$

Solution: $P = \frac{1}{0.2} - \frac{1}{0.3} = \frac{5}{3}\,\text{D} > 0$; $F = 0.6\,\text{m} = 60\,\text{cm}$, *converging.*

Answer: (A)

Q6 · JEE Main level · Numerical

A prism of angle 60° (in air) gives a minimum deviation of 30°. Its refractive index?

Solution: $n = \frac{\sin 45^\circ}{\sin 30^\circ} = \sqrt{2} \approx 1.41$.

Answer: ≈ 1.41

Q7 · JEE Main level · Numerical

Compound microscope: $f_o = 1\,\text{cm}$, $f_e = 5\,\text{cm}$, object $1.1\,\text{cm}$ from the objective, final image at the near point. $|M|$?

Solution: $\frac{1}{v_o} = \frac{1}{1} - \frac{1}{1.1} = \frac{1}{11}$, $v_o = 11\,\text{cm}$, $|m_o| = 10$; $m_e = 1 + \frac{25}{5} = 6$; $|M| = 10 \times 6 = 60$.

Answer: 60

Q8 · JEE Main level · MCQ

Astronomical telescope, $f_o = 100\,\text{cm}$, $f_e = 5\,\text{cm}$, normal adjustment. M and length: (A) 20, $105\,\text{cm}$ (B) 24, $105\,\text{cm}$ (C) 20, $95\,\text{cm}$ (D) 24, $104.2\,\text{cm}$

Solution: $M = \frac{100}{5} = 20$, $L = 100 + 5 = 105\,\text{cm}$ *(image at D would give 24: trap).*

Answer: (A)

Questions: JEE Advanced Level (I)

paraxial rays, thin lenses in air, Cartesian sign convention

A1 · JEE Advanced level · Numerical

A thin plano-convex lens ($n = 1.5$, $R = 20\,\text{cm}$) has its PLANE face silvered. An object is placed on its axis $30\,\text{cm}$ in front of it. How far (in cm) from the lens is the image?

Solution: $P_{eq} = 2P_L + P_M = 2 \cdot \frac{0.5}{0.2} + 0 = 5\,\text{D}$, $f_{eq} = -20\,\text{cm}$ (concave mirror). $\frac{1}{v} = -\frac{1}{20} + \frac{1}{30} = -\frac{1}{60}$; $v = -60\,\text{cm}$, real, $m = -2$.

Answer: 60

A2 · JEE Advanced level · Multi-correct

A glass sphere ($n = 1.5$) of radius $10\,\text{cm}$ is in air. A point object is on a diameter line, $30\,\text{cm}$ from the near surface. Then: (A) the near surface alone images it $90\,\text{cm}$ beyond that surface (B) the final image is real, $14\,\text{cm}$ beyond the far surface (C) the final image is virtual (D) the first image lies inside the sphere

Solution: Surface 1: $\frac{1.5}{v} - \frac{1}{-30} = \frac{0.5}{10}$, $v = 90\,\text{cm}$ (outside the $20\,\text{cm}$ sphere). Surface 2: $u = 90 - 20 = +70\,\text{cm}$, $\frac{1}{v} - \frac{1.5}{70} = \frac{-0.5}{-10}$, $v = 14\,\text{cm}$ (real).

Answer: (A), (B)

Golden points

- ★ Silvered lens: light crosses the lens twice, $P_{eq} = 2P_L + P_M$; then use the MIRROR formula with $f_{eq} = -1/P_{eq}$.
- ★ Two surfaces: the first image is the object for the second; re-measure from the new vertex ($90 - 20 = 70\,\text{cm}$, a virtual object).

Questions: JEE Advanced Level (2)

paraxial rays, thin lenses in air

A3 · JEE Advanced level · Multi-correct

A convex lens ($f = 20\,\text{cm}$) has a plane mirror $20\,\text{cm}$ behind it, perpendicular to the axis. An object is $60\,\text{cm}$ in front of the lens. Then: (A) the lens first images it $30\,\text{cm}$ behind the lens (B) the final image is $20\,\text{cm}$ from the lens, at the mirror (C) the final image is real (D) the final image is inverted and of the same size

Solution: Lens: $\frac{1}{v} = \frac{1}{20} - \frac{1}{60}$, $v = 30$, $m_1 = -0.5$. That is $10\,\text{cm}$ behind the mirror; the mirror images it $10\,\text{cm}$ in front (at $10\,\text{cm}$ from the lens), $m_2 = 1$. Light now goes right to left: $u = -10$, $\frac{1}{v} = \frac{1}{20} - \frac{1}{10}$, $v = -20$: virtual, $20\,\text{cm}$ from the lens on the mirror side; $m_3 = 2$. Total $m = -1$.

Answer: (A), (B), (D)

A4 · JEE Advanced level · Paragraph

An optical fibre has core index 1.5 and cladding index 1.2. (i) Largest angle of incidence at the end face (from air) for which light is guided? (ii) The same if the end face is in water ($4/3$)?

Solution: (i) $\sin\theta_a = \sqrt{1.5^2 - 1.2^2} = 0.9$, $\theta_a = 64.2^\circ$. (ii) $\frac{4}{3}\sin\theta_a = 0.9$, $\sin\theta_a = 0.675$, $\theta_a = 42.5^\circ$.

Answer: (i) 64.2° (ii) 42.5°

Golden points

- ★ After a mirror the light travels right to left: take $+$ along the NEW direction of the light before using the lens formula again.
- ★ Fibre: $n_0 \sin\theta_a = \sqrt{n_1^2 - n_2^2}$ is fixed, so a denser outside medium gives a smaller acceptance angle.

Questions: JEE Advanced Level (3)

real object, paraxial rays, lens or mirror in air

AS · JEE Advanced level · Matching

List I: (P) convex mirror, object anywhere (Q) concave mirror, object between F and P (R) convex lens, object at $2F_1$ (S) convex lens, object at F_1. List II: (1) virtual, erect, diminished (2) virtual, erect, magnified (3) real, inverted, same size (4) image at infinity

Solution: (P) always virtual, erect, diminished; (Q) $|u| < |f|$ gives $v > 0$, $m > 1$; (R) $v = 2f$, $m = -1$; (S) $\frac{1}{v} = \frac{1}{f} - \frac{1}{f} = 0$.

Answer: P-1, Q-2, R-3, S-4

A6 · JEE Advanced level · Numerical

An object on the axis of a concave mirror ($|f| = 20\,\mathrm{cm}$), $30\,\mathrm{cm}$ away, moves towards the mirror at $2\,\mathrm{cm\,s^{-1}}$ (mirror at rest). Speed of the image in $\mathrm{cm\,s^{-1}}$?

Solution: $\frac{1}{v} = -\frac{1}{20} + \frac{1}{30}$, $v = -60\,\mathrm{cm}$, $m = -2$.

$v_I = -m^2 v_O = -4 \times (+2) = -8\,\mathrm{cm\,s^{-1}}$: the image moves AWAY from the mirror at $8\,\mathrm{cm\,s^{-1}}$.

Answer: 8

Golden points

- ★ Image velocity: mirror $v_I = -m^2 v_O$, lens $v_I = +m^2 v_O$ (along the axis).
- ★ Nature from the sign of m (single mirror or lens, real object): $m < 0$ real and inverted, $m > 0$ virtual and erect.